Chapter 1 Riemann Surfaces

1.1 Definitions

Definition 1.1.0.1 A **Riemann surface** is one-dimensional complex=conneected 1-dimensional complex analytic manifold.

That is, equipped with maximal charts $\{(U_\alpha, z_\alpha)\}_{\alpha \in A}$ on M

$$z_\alpha : U_\alpha \to \mathbb{C}$$

Then transition functions are holomorphic

$$f_{\alpha\beta} = z_\alpha \circ z_\beta^{-1} :; U_\alpha \cap U_\beta = \varnothing$$

The charts are called **transition functions**.

Notice that: The set of holomorphic functions forms a pseudogroup under composition. Taking

$$\mathcal{H}(X) = \{f : U \to V | f \text{ is biholomorphism}; U, V \subseteq X \text{ are open}\}$$

Then,

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Definition 1.1.0.2 Classically, a compact Riemann surface is called **closed**, elsewise is **open**.

◇ **Lemma 1.1.0.3** ...

∘ **Proof** ... □

1.1.1 Holomorphic Mapping

1.2 Topology of Riemann Surface

1.2.1 Fundamental Groups

1.2.2 Covering Manifolds

1.2.3 Riemann-Hurwitz Relation

▶ ***Theorem*** We have

$$g = n(\gamma - 1) + 1 + B/2$$

1.3 Differential Forms

1.4 Integration Forms

Chapter 2 Analytic Continuation, Covering Surfaces and Algebraic Functions

2.1 Analytic Continuation

Definition 2.1.0.1

- (Function Element) Let M, N be fixed Riemann surfaces. A **function element** is pair

$$(f, D)$$

  where:

  - D is a connected, open nonempty subset of M;
  - $f : D \to N$ is analytic.

- Two elements $(f_1, D_1), (f_2, D_2)$ are **direct analytic continuation** of each other if

$$D_1 \cap D_2 \neq \emptyset \qquad f_1|D_1 \cap D_2 = f_2|D_1 \cap D_2$$

  Use the following notion $(f_1, D_1) \simeq (f_2, D_2)$.

Definition 2.1.0.2 (Analytic Continuations)

- We say (f, D), $(g, \tilde{D})$ are **analytic continuation** if there exists function elements (f_j, D_j); $1 \leq j \leq J$:

$$(f_1, D_1) = (f, D) \qquad (f_J, D_J) = (g, \tilde{J}) \qquad (f_j, D_j) \simeq (f_{j+1}, D_{j+1})$$

- A **compelete analytic continuation** is an equivalence class of suc (f, D). By analytic continuation.

◇ **Lemma 2.1.0.3** (Analytic Continuation⇒ Path) Suppose $(f, D), (g, \tilde{D})$ is an analytic continuation. Then, there exists a path

$$\gamma : [0, 1] \to M$$

Connecting the points of D, with following property: Exists

$$0 < t_1 < \cdots < t_m = 1$$

where: Exists $B(\gamma(t_i), r_i)$

∘ **Proof** ... □

★ **Corollary 2.1.0.4** ...

∘ **Proof** ... □

2.2 The Unramified Riemann Surface of An Analytic Germ

2.3 The Ramified Riemann Surface of An Analytic Germ

2.4 Algebraic Germs and Functions

2.5 title

Chapter 3 Stein Manifold

Chapter 4 Compact Riemann Surface

4.1 Existence Theorem